

OpenAI API Key Integration Guide

AI File Intelligence Bot | RAG + OCR + React + Python

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01 What Is the OpenAI API Key?

An OpenAI API key is a private credential that allows your application to call OpenAI's language models (like GPT-3.5 Turbo or GPT-4o) over the internet. Think of it as a password your server uses to talk to OpenAI's brain.

In your AI File Intelligence Bot, the API key is used in **llm.py**. When a user asks a question, the system retrieves relevant document chunks from ChromaDB, then sends those chunks + the question to OpenAI to generate a smart, synthesized answer.

Without the key →

The system still works! It runs in **context-only mode** — it retrieves chunks and returns the most relevant text directly, without AI synthesis. No GPT, no cost.

With the key →

Full AI mode. OpenAI reads the retrieved chunks and writes a clean, human-readable, synthesized answer for the user.

02 One Key or One Per Team Member?

This is one of the most common questions. Here is the definitive answer for your setup:

ANSWER	ONE API KEY — Shared by the whole team
WHY?	The key lives in the .env file on the SERVER (your backend machine). All team members connect to the server.
WHO MANAGES IT?	Jameela (Admin) — only the person who controls the server needs the key.
DO TEAM MEMBERS NEED ACCOUNTS?	NO. Team members only need access to the web UI. They do not need an OpenAI account.

■ ■ Important: Each API call consumes tokens billed to your OpenAI account. Monitor usage at platform.openai.com to avoid unexpected charges.

03 How to Get Your API Key

	Go to https://platform.openai.com and sign in (or create a free account).
	Click your profile icon (top-right) → click 'API Keys' in the left sidebar.
	Click '+ Create new secret key' → give it a name like 'AI-FileBot-Server'.
	COPY the key immediately. It starts with sk-... and will NOT be shown again.
	Store it securely. Do NOT share it via WhatsApp, email, or Slack.
	Add billing: go to Settings → Billing → Add payment method. Set a monthly spend limit (e.g. \$10) to avoid surprises.

Tip: You can create multiple keys (one per project). If a key is ever compromised, delete only that key without affecting other projects.

04 How to Add the Key to This System

Your system reads the API key from a file called `.env` in the project root. This file is never committed to Git (it is in `.gitignore`) so it stays private on your machine.

4.1 Locate the .env file

The project root already contains a file called **.env.example**. If you have not already created the real `.env` file, do this:

```
# On Windows (Command Prompt)
copy .env.example .env

# On Mac / Linux (Terminal)
cp .env.example .env
```

4.2 Open .env and add your key

Open the `.env` file in any text editor (Notepad, VS Code, etc.) and replace the placeholder:

[illegible]

4.3 Restart the backend server

```
# Windows

start_windows.bat

# Mac / Linux

bash start.sh

# Or directly

cd backend

uvicorn main:app --reload --host 0.0.0.0 --port 8000
```

4.4 Verify the key is working

After restart, open the chat UI, upload any document, and ask a question. If you see **mode: openai** in the API response (check browser DevTools → Network → /query), the key is working. If you see **mode: context-only**, the key was not picked up — check the .env file for typos and restart.

You can also test the key directly in Python:

```
from openai import OpenAI
client = OpenAI(api_key='sk-your-key-here')
r = client.chat.completions.create(
    model='gpt-3.5-turbo',
    messages=[{'role': 'user', 'content': 'Say hello'}],
    max_tokens=10
)
print(r.choices[0].message.content) # Should print: Hello!
```

05 Choosing the Right Model

The `LLM_MODEL` setting in `.env` controls which OpenAI model your system uses. Each model has different speed, quality, and cost:

Model	Speed	Quality	Cost per 1K tokens	Best For
gpt-3.5-turbo	Very Fast	Good	~\$0.0015 input ~\$0.002 output	Daily use, testing, high volume
gpt-4o-mini	Fast	Very Good	~\$0.00015 input ~\$0.0006 output	Best value — recommended
gpt-4o	Medium	Excellent	~\$0.005 input ~\$0.015 output	Complex reasoning, critical tasks
gpt-4-turbo	Slower	Excellent	~\$0.01 input ~\$0.03 output	Long documents, deep analysis

■ *Recommendation: Start with **gpt-4o-mini** — it offers the best balance of quality and cost for a RAG system where most of the intelligence comes from retrieved context, not the model.*

To change the model, update `.env`:

```
LLM_MODEL=gpt-4o-mini # Recommended — best value
# LLM_MODEL=gpt-3.5-turbo # Faster and cheaper
# LLM_MODEL=gpt-4o # Highest quality
```

06 Cost Estimation

OpenAI charges per token (roughly 1 token = 0.75 words). Your RAG system sends retrieved context + the user's question in each call. Here is a rough estimate:

Usage Level	Queries/Day	Est. Tokens/Query	Daily Cost	Monthly Cost
Light (UAT testing)	~20	~1,500	~\$0.05	~\$1.50
Moderate (small team)	~100	~1,500	~\$0.25	~\$7.50
Active (full team)	~500	~2,000	~\$1.50	~\$45

Estimates based on gpt-4o-mini pricing. Actual costs vary. Always set a monthly spend limit at [platform.openai.com](https://platform.openai.com/settings/limits) → Settings → Limits.

07 Security Rules for the API Key

■ DO	■ DO NOT
Store the key ONLY in the <code>.env</code> file on the server	Paste the key in WhatsApp, Teams, or email
Add <code>.env</code> to <code>.gitignore</code> (already done in your project)	Commit the <code>.env</code> file to GitHub or any Git repo

Set a monthly spend limit on OpenAI dashboard	Share the key with team members directly
Rotate (regenerate) the key every 3-6 months	Hardcode the key inside any Python or JS file
Use environment variables — never hardcode	Leave the key visible in browser console logs
Restrict key usage to specific IPs if possible (OpenAI Dashboard)	Use the same key for multiple projects

If you suspect your key has been exposed: go to platform.openai.com → API Keys → Delete the compromised key immediately → Create a new one → Update .env → Restart server.

08 Troubleshooting

Problem	Likely Cause	Fix
mode: context-only in response	.env not loaded or key missing	Check .env exists. Restart backend. Print OPENAI_API_KEY in Python
AuthenticationError: Incorrect API key	Wrong or expired key in .env	Re-copy key from platform.openai.com. Remove all spaces. No quotes
RateLimitError: quota exceeded	No billing set up or monthly limit hit	Add payment method at platform.openai.com → Billing. Raise or reset s
openai.APIConnectionError	No internet or firewall blocking	Check server internet access. Ensure outbound HTTPS to api.openai.co
model not found / 404 error	Wrong model name in LLM_MODEL	Check spelling. Valid: gpt-3.5-turbo, gpt-4o-mini, gpt-4o, gpt-4-turbo.
Slow responses (>10 sec)	Using gpt-4 with large context	Switch to gpt-4o-mini or reduce TOP_K_RETRIEVAL in .env from 5 to 3

09 Quick Reference Cheat Sheet

Complete .env configuration

```
# █ Copy to .env in project root ████████████████████████████████████████  
  
# LLM – Your OpenAI key (admin only, never share)  
  
OPENAI_API_KEY=sk-proj-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx  
  
# Model (gpt-4o-mini recommended)  
  
LLM_MODEL=gpt-4o-mini  
  
LLM_MAX_TOKENS=1024  
  
LLM_TEMPERATURE=0.2  
  
# Embeddings (local, FREE – no key needed)  
  
EMBEDDING_MODEL=all-MiniLM-L6-v2  
  
# RAG settings  
  
CHUNK_SIZE=800  
  
CHUNK_OVERLAP=100  
  
TOP_K_RETRIEVAL=5  
  
# ChromaDB  
  
CHROMA_COLLECTION=file_intelligence  
  
# Server  
  
HOST=0.0.0.0  
  
PORT=8000  
  
CORS_ORIGINS=*
```

Key facts to remember

Question	Answer
Who needs the API key?	Only Jameela (admin / server owner)
How many keys needed?	ONE — shared via the server .env file
Do team members need OpenAI accounts?	NO
Where is the key stored?	In .env file on the server (never in code)
What happens without the key?	System runs in context-only mode (still works)
How to verify the key works?	Check API response for mode: openai
Recommended model?	gpt-4o-mini (best value)
How to control costs?	Set spend limit at platform.openai.com → Billing

How to rotate the key?

Delete old key → create new → update .env → restart

For questions or issues, refer to: <https://platform.openai.com/docs> | Prepared by AI File Intelligence Bot Project | Admin: Jameela